

A random Schauder–Tychonoff fixed point theorem

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The open problem

Problem 5.1 of T. Guo, Y. Wang, H.-K. Xu, G. X. Yuan and G. Chen, *The noncompact Schauder fixed point theorem in random normed modules and its applications*, arXiv:2104.11095, asks the following. Let $(E, \mathcal{P})$ be a random locally convex module and let $G \subset E$ be a stably compact L^0 -convex set. Must every σ -stable self-map of G which is either $\mathcal{T}_{\varepsilon, \lambda}(\mathcal{P})$ -continuous or $\mathcal{T}_c(\mathcal{P})$ -continuous have a fixed point?

The source proves this for random normed modules. A 2024 theorem of Q. Tu, X. Mu and T. Guo proves the random Markov–Kakutani theorem in random locally convex modules, but only for L^0 -affine maps. We give an affirmative answer for the arbitrary nonlinear maps in Problem 5.1.

We use the standard notation of those papers. Thus $\mathcal{P}_{cc}$ is the σ -stable hull of $\mathcal{P}$: its members are countable concatenations of finite maxima of seminorms in $\mathcal{P}$. Stable compactness may equivalently be taken with respect to $\mathcal{T}_c(\mathcal{P}_{cc})$. As in the cited framework, modules are regular and all equalities and inequalities are modulo null sets.

Full resolution

Theorem 1 (Random Schauder–Tychonoff theorem). *Let $(E, \mathcal{P})$ be a random locally convex module over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ with base $(\Omega, \mathcal{F}, P)$, and let G be a nonempty stably compact L^0 -convex subset of E . If $T : G \rightarrow G$ is σ -stable and is either*

(i) $\mathcal{T}_{\varepsilon, \lambda}(\mathcal{P})$ -continuous, or

(ii) $\mathcal{T}_c(\mathcal{P})$ -continuous,

then T has a fixed point.

The proof has two layers. Stable compactness first produces a piecewise-finite net for one seminorm. The random Brouwer theorem then gives an approximate fixed point for that seminorm. Finally, stable compactness is used a second time on the family of all closed approximate-fixed-point sets.

Stable finite nets

For $q \in \mathcal{P}_{cc}$, $r \in L_{++}^0(\mathcal{F})$ and $x \in G$, put

$$B_G(x; q, r) = \{y \in G : q(y - x) < r \text{ on } \Omega\}.$$

Lemma 1 (Piecewise-finite seminorm net). *For every $q \in \mathcal{P}_{cc}$ and $r \in L_{++}^0(\mathcal{F})$ there are a countable partition (A_n) of Ω and finite sets*

$$D_n = \{x_{n1}, \dots, x_{nk_n}\} \subset G$$

such that, for every $z \in G$ and every n ,

$$\tilde{I}_{A_n} \min_{1 \leq i \leq k_n} q(z - x_{ni}) < \tilde{I}_{A_n} r. \tag{1}$$

Proof. The family $\mathcal{O} = \{B_G(x; q, r) : x \in G\}$ is a σ -stable family of σ -stable $\mathcal{T}_c(\mathcal{P}_{cc})$ -open sets. Indeed, locality of q gives

$$\sum_m \tilde{I}_{C_m} B_G(x_m; q, r) = B_G\left(\sum_m \tilde{I}_{C_m} x_m; q, r\right)$$

for every measurable partition (C_m) . It covers G . The stable open-cover characterization of stable compactness therefore supplies a stable finite subcover. By definition, it has the form

$$\sum_n \tilde{I}_{A_n} \sigma(\mathcal{O}_n), \quad \mathcal{O}_n = \{B_G(x_{ni}; q, r) : 1 \leq i \leq k_n\}.$$

If $z \in G$, membership in this subcover gives, on A_n , a measurable partition among the finitely many balls in $\mathcal{O}_n$. Equivalently, the pointwise minimum in (1) is strictly smaller than r . $\square$

One-seminorm approximate fixed points

Lemma 2. *Under the hypotheses of Theorem 1, for every $q \in \mathcal{P}_{cc}$ and $r \in L_{++}^0(\mathcal{F})$ there exists $x \in G$ such that*

$$q(x - Tx) < r \quad \text{on } \Omega. \quad (2)$$

Proof. Choose (A_n) and D_n from Lemma 1. Fix n and write $k = k_n$, $x_i = x_{ni}$ and $A = A_n$. On the standard random simplex

$$\Delta_k = \{\alpha = (\alpha_1, \dots, \alpha_k) \in (L_+^0(\mathcal{F}))^k : \sum_i \alpha_i = 1\},$$

define $h(\alpha) = \sum_i \alpha_i x_i \in G$. For $z \in G$, set

$$\begin{aligned} u_1(z) &= \tilde{I}_A(r - q(z - x_1))_+ + \tilde{I}_{A^c}, \\ u_i(z) &= \tilde{I}_A(r - q(z - x_i))_+, \quad 2 \leq i \leq k, \end{aligned}$$

and $U(z) = \sum_i u_i(z)$. By (1), $U(z) > 0$ on A ; by construction $U(z) = 1$ on A^c . Hence

$$\Phi(\alpha) = \left(\frac{u_1(T(h(\alpha)))}{U(T(h(\alpha)))}, \dots, \frac{u_k(T(h(\alpha)))}{U(T(h(\alpha)))} \right) \quad (3)$$

is a well-defined self-map of Δ_k .

The map Φ is σ -stable. It is also continuous in the topology needed for the random Brouwer theorem. In case (i), $\mathcal{T}_{\varepsilon, \lambda}(\mathcal{P}) = \mathcal{T}_{\varepsilon, \lambda}(\mathcal{P}_{cc})$, and q , h , positive part, addition, multiplication and inversion on L_{++}^0 are continuous for the corresponding (ε, λ) topologies. In case (ii), the 2024 continuity lemma says that a σ -stable $\mathcal{T}_c(\mathcal{P})$ -continuous map is $\mathcal{T}_c(\mathcal{P}_{cc})$ -continuous; the same operations are $\mathcal{T}_c$ -continuous. After restriction to the finite-rank random simplex, either continuity hypothesis gives the random sequential continuity used by the source's complete random Brouwer theorem.

Consequently, Φ has a fixed point $\alpha^n \in \Delta_k$. Put $y_n = h(\alpha^n)$. On A_n , the fixed-point identity says

$$y_n = \sum_{i=1}^{k_n} \frac{(r - q(Ty_n - x_{ni}))_+}{\sum_j (r - q(Ty_n - x_{nj}))_+} x_{ni}.$$

Only indices satisfying $q(Ty_n - x_{ni}) < r$ receive positive weight. The seminorm inequality therefore gives

$$\tilde{I}_{A_n} q(y_n - Ty_n) < \tilde{I}_{A_n} r. \quad (4)$$

Finally concatenate $x = \sum_n \tilde{I}_{A_n} y_n$. The set G and the map T are σ -stable, while q is local. Thus $x \in G$ and (4) concatenates to (2). $\square$

This is precisely the source paper's random Schauder projection, but run on the finitely many centers available on each stable stratum and applied to one seminorm at a time. The coefficient-simplex formulation avoids quotienting by $\ker q$; quotienting would be invalid because a nonlinear T need not descend to that quotient.

From approximate to exact

For $q \in \mathcal{P}_{cc}$ and $r \in L_{++}^0(\mathcal{F})$, define

$$C(q, r) = \{x \in G : q(x - Tx) \leq r\}.$$

Lemma 2 shows that every $C(q, r)$ is nonempty. Each is σ -stable. It is also $\mathcal{T}_c(\mathcal{P}_{cc})$ -closed: in case (ii) this follows from $\mathcal{T}_c(\mathcal{P}_{cc})$ -continuity; in case (i) it is first $\mathcal{T}_{\varepsilon, \lambda}(\mathcal{P}_{cc})$ -closed, and the closure theorem for σ -stable sets says that $\mathcal{T}_{\varepsilon, \lambda}(\mathcal{P})$, $\mathcal{T}_c(\mathcal{P})$ and $\mathcal{T}_c(\mathcal{P}_{cc})$ have the same σ -stable closed sets.

The family

$$\mathcal{C} = \{C(q, r) : q \in \mathcal{P}_{cc}, r \in L_{++}^0(\mathcal{F})\} \quad (5)$$

is itself σ -stable. To see this, let (B_m) be a measurable partition and let $C(q_m, r_m) \in \mathcal{C}$. Then

$$q = \sum_m \tilde{I}_{B_m} q_m \in \mathcal{P}_{cc}, \quad r = \sum_m \tilde{I}_{B_m} r_m \in L_{++}^0(\mathcal{F}),$$

and locality of T and the seminorms gives

$$\sum_m \tilde{I}_{B_m} C(q_m, r_m) = C(q, r). \quad (6)$$

For the reverse inclusion in (6), use a filler point in each nonempty $C(q_m, r_m)$ off B_m and concatenate; this turns the local inequality for a point of $C(q, r)$ into a global member of each $C(q_m, r_m)$.

The family has the finite-intersection property. For $C(q_i, r_i)$, $1 \leq i \leq N$, take

$$q = \max_{1 \leq i \leq N} q_i \in \mathcal{P}_{cc}, \quad r = \min_{1 \leq i \leq N} r_i \in L_{++}^0(\mathcal{F}).$$

The assertion $q \in \mathcal{P}_{cc}$ follows by taking a common countable refinement of the representations of the q_i ; finite maxima remain finite maxima. Then

$$\emptyset \neq C(q, r) \subseteq \bigcap_{i=1}^N C(q_i, r_i).$$

Stable compactness and its closed-set finite-intersection characterization now imply $\bigcap \mathcal{C} \neq \emptyset$. Choose x in this intersection. For every $p \in \mathcal{P}$ and every positive integer m ,

$$p(x - Tx) \leq \frac{1}{m}.$$

Hence $p(x - Tx) = 0$ for all $p \in \mathcal{P}$. Since $\mathcal{P}$ separates points, $x = Tx$. This completes the proof of Theorem 1.

Why the proof closes both continuity cases

The weaker-topology case is not obtained by claiming that every $\mathcal{T}_{\varepsilon, \lambda}$ -continuous map on G is globally $\mathcal{T}_c$ -continuous. Instead, the only map to which Brouwer is applied is the finite-dimensional coefficient map (3). If T is $\mathcal{T}_{\varepsilon, \lambda}$ -continuous, then (3) is $\mathcal{T}_{\varepsilon, \lambda}$ -continuous on a random Euclidean simplex, exactly a case covered by the source's random Brouwer theorem. If T is $\mathcal{T}_c$ -continuous, then (3) is $\mathcal{T}_c$ -continuous there and the source's equivalence with random sequential continuity applies. The final closed-set argument uses the equality of closures only for the σ -stable sets $C(q, r)$, where that equality is valid.

Literature and novelty audit

A bounded search through 13 August 2026 used the arXiv id, exact problem language, title, authors, and combinations of “stably compact”, “random locally convex module”, “Schauder–Tychonoff” and “fixed point”. The closest later primary result found was the 2024 random Markov–Kakutani theorem, which assumes L^0 -affinity. The 2025 random Kakutani theorem is again restricted to random normed modules. A March 2026 survey by several of the same authors lists those results and describes the stable-compact topological theory as still under development; it does not state the nonlinear theorem above. No posted full resolution of Problem 5.1 was located.

References

- T. Guo, Y. Wang, H.-K. Xu, G. X. Yuan and G. Chen, *The noncompact Schauder fixed point theorem in random normed modules and its applications*, Math. Ann. 391 (2025), 3863–3911; arXiv:2104.11095. In particular: the complete random Brouwer theorem, the random Schauder projection (Lemmas 3.7–3.9), and Problem 5.1.
- Q. Tu, X. Mu and T. Guo, *The random Markov–Kakutani fixed point theorem in a random locally convex module*, New York J. Math. 30 (2024), 1196–1219. In particular: Propositions 2.8, 2.14 and 2.15 and Lemma 3.8.
- Q. Tu, X. Mu, T. Guo, G. Yang and Y. Sun, *The random Kakutani fixed point theorem in random normed modules*, arXiv:2509.15649 (2025).
- T. Guo, Q. Tu, X. Mu and Y. Sun, *Survey of Metric fixed point theory in random functional analysis*, arXiv:2603.27963 (2026).

Human-review recommendation. Review as a full resolution of Problem 5.1. The proof-critical checks are: (1) extracting (1) from a stable finite subcover; (2) continuity and σ -stability of (3); (3) the local fixed-point identity and concatenation in Lemma 2; and (4) stable closedness and the finite-intersection argument for (5).